

Compte-rendu du TP1 en R203

Serveur web Apache2 & Serveur Web Nginx

Installation du serveur apache

- sudo apt update (MAJ des paquets)
- sudo apt install apache2 (Installation des paquets apache2)
- systemctl start apache2 (démarrer les services apache)
- systemctl status apache2 (vérifier l'état du serveur « activé/désactiver »)

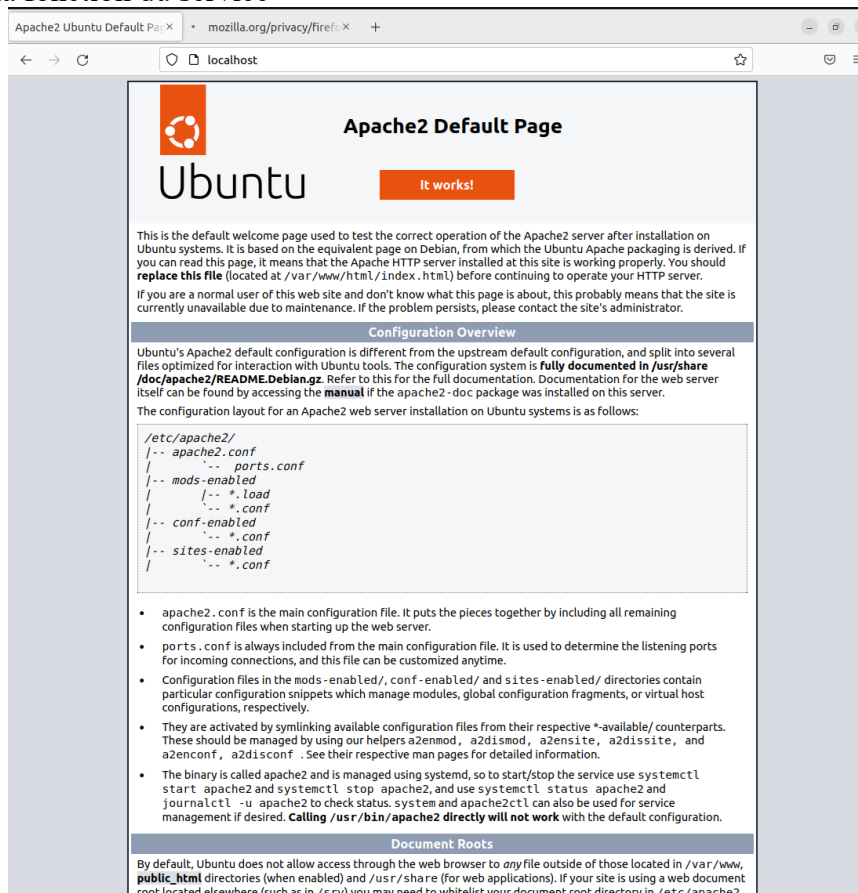
Les fichiers de configuration du serveur apache2 se trouvent dans le répertoire /etc/apache2

*

```
administrateur@rt-mv:~$ systemctl start apache2
administrateur@rt-mv:~$ systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/lib/systemd/system/apache2.service; enabled; vendor prese
   Active: active (running) since Mon 2026-03-30 08:37:15 CEST; 1min 10s ago
     Docs: https://httpd.apache.org/docs/2.4/
    Main PID: 3407 (apache2)
      Tasks: 55 (limit: 2878)
     Memory: 5.0M
        CPU: 30ms
    CGroup: /system.slice/apache2.service
            └─3407 /usr/sbin/apache2 -k start
              └─3408 /usr/sbin/apache2 -k start
                └─3409 /usr/sbin/apache2 -k start

mars 30 08:37:15 rt-mv systemd[1]: Starting The Apache HTTP Server...
mars 30 08:37:15 rt-mv apache2[3406]: AH00558: apache2: Could not reliably de
mars 30 08:37:15 rt-mv systemd[1]: Started The Apache HTTP Server.
lines 1-16/16 (END)
```

Vérification de la fonction du service



Apache2 Ubuntu Default Page

It works!

This is the default welcome page used to test the correct operation of the Apache2 server after installation on Ubuntu systems. It is based on the equivalent page on Debian, from which the Ubuntu Apache packaging is derived. If you can read this page, it means that the Apache HTTP server installed at this site is working properly. You should **replace this file** (located at `/var/www/html/index.html`) before continuing to operate your HTTP server.

If you are a normal user of this web site and don't know what this page is about, this probably means that the site is currently unavailable due to maintenance. If the problem persists, please contact the site's administrator.

Configuration Overview

Ubuntu's Apache2 default configuration is different from the upstream default configuration, and split into several files optimized for interaction with Ubuntu tools. The configuration system is **fully documented in `/usr/share/doc/apache2/README.Debian.gz`**. Refer to this for the full documentation. Documentation for the web server itself can be found by accessing the **manual** if the `apache2-doc` package was installed on this server.

The configuration layout for an Apache2 web server installation on Ubuntu systems is as follows:

```
/etc/apache2/
|-- apache2.conf
|   |-- ports.conf
|-- mods-enabled
|   |-- *.load
|   |-- *.conf
|-- conf-enabled
|   |-- *.conf
|-- sites-enabled
|   |-- *.conf
```

- `apache2.conf` is the main configuration file. It puts the pieces together by including all remaining configuration files when starting up the web server.
- `ports.conf` is always included from the main configuration file. It is used to determine the listening ports for incoming connections, and this file can be customized anytime.
- Configuration files in the `mods-enabled/`, `conf-enabled/` and `sites-enabled/` directories contain particular configuration snippets which manage modules, global configuration fragments, or virtual host configurations, respectively.
- They are activated by symlinking available configuration files from their respective `*-available/` counterparts. These should be managed by using our helpers `a2enmod`, `a2dismod`, `a2ensite`, `a2dissite`, and `a2enconf`, `a2disconf`. See their respective man pages for detailed information.
- The binary is called `apache2` and is managed using `systemd`, so to start/stop the service use `systemctl start apache2` and `systemctl stop apache2`, and use `systemctl status apache2` and `journalctl -u apache2` to check status. `systemd` and `apache2ctl` can also be used for service management if desired. **Calling `/usr/bin/apache2` directly will not work** with the default configuration.

Document Roots

By default, Ubuntu does not allow access through the web browser to any file outside of those located in `/var/www/public_html` directories (when enabled) and `/usr/share` (for web applications). If your site is using a web document root located elsewhere (such as in `/srv`) you may need to whitelist your document root directory in `/etc/apache2`

Extrait de
d'un fichier de

configuration
configuration

```

administrateur@rt-mv:/etc/apache2$ cat apache2.conf
# This is the main Apache server configuration file.  It contains the
# configuration directives that give the server its instructions.
# See http://httpd.apache.org/docs/2.4/ for detailed information about
# the directives and /usr/share/doc/apache2/README.Debian about Debian specific
# hints.
#
#
# Summary of how the Apache 2 configuration works in Debian:
# The Apache 2 web server configuration in Debian is quite different to
# upstream's suggested way to configure the web server. This is because Debian's
# default Apache2 installation attempts to make adding and removing modules,
# virtual hosts, and extra configuration directives as flexible as possible, in
# order to make automating the changes and administering the server as easy as
# possible.

# It is split into several files forming the configuration hierarchy outlined
# below, all located in the /etc/apache2/ directory:
#
#       /etc/apache2/
#       |-- apache2.conf
#       |   `-- ports.conf

```

Configuration de base d'apache2

```

root@rt-mv:/etc/apache2/sites-available# cat 000-default.conf
<VirtualHost *:80>
    # The ServerName directive sets the request scheme, hostname and port that
    # the server uses to identify itself. This is used when creating
    # redirection URLs. In the context of virtual hosts, the ServerName
    # specifies what hostname must appear in the request's Host: header to
    # match this virtual host. For the default virtual host (this file) this
    # value is not decisive as it is used as a last resort host regardless.
    # However, you must set it for any further virtual host explicitly.
    #ServerName www.example.com

    ServerAdmin webmaster@localhost
    DocumentRoot /var/www/html

    # Available loglevels: trace8, ..., trace1, debug, info, notice, warn,
    # error, crit, alert, emerg.
    # It is also possible to configure the loglevel for particular
    # modules, e.g.
    #LogLevel info ssl:warn

    ErrorLog ${APACHE_LOG_DIR}/error.log
    CustomLog ${APACHE_LOG_DIR}/access.log combined

    # For most configuration files from conf-available/, which are
    # enabled or disabled at a global level, it is possible to
    # include a line for only one particular virtual host. For example the
    # following line enables the CGI configuration for this host only
    # after it has been globally disabled with "a2disconf".
    #Include conf-available/serve-cgi-bin.conf
</VirtualHost>

```

*

Pages protégées

Activer les outils nécessaires

- sudo a2enmod authz_groupfile (activation des modules)
- sudo systemctl restart apache2

Créer les dossiers et les utilisateurs :

- sudo mkdir /var/www/html/protège (le dossier)

Le fichier mdp (placé en dehors du web pour la sécurité)

- sudo htpasswd -c /etc/apache2/.htpasswd user1
- sudo htpasswd /etc/apache2/.htpasswd user2

Créer le fichier groupe

- nano /etc/apache2/RT1

contenu :

RT1: user1 user2

Configurer l'accès dans le VirtualHost (fichier 000-default.conf)

```
sudo nano 0.2 /etc/apache2/sites-available/000-default.conf
<VirtualHost *:80>
    # The ServerName directive sets the request scheme, hostname and port that
    # the server uses to identify itself. This is used when creating
    # redirection URLs. In the context of virtual hosts, the ServerName
    # specifies what hostname must appear in the request's Host: header to
    # match this virtual host. For the default virtual host (this file) this
    # value is not decisive as it is used as a last resort host regardless.
    # However, you must set it for any further virtual host explicitly.
    #ServerName www.example.com

    ServerAdmin webmaster@localhost
    DocumentRoot /var/www/html

    # Available loglevels: trace8, ..., trace1, debug, info, notice, warn,
    # error, crit, alert, emerg.
    # It is also possible to configure the loglevel for particular
    # modules, e.g.
    #LogLevel info ssl:warn

    ErrorLog ${APACHE_LOG_DIR}/error.log
    CustomLog ${APACHE_LOG_DIR}/access.log combined

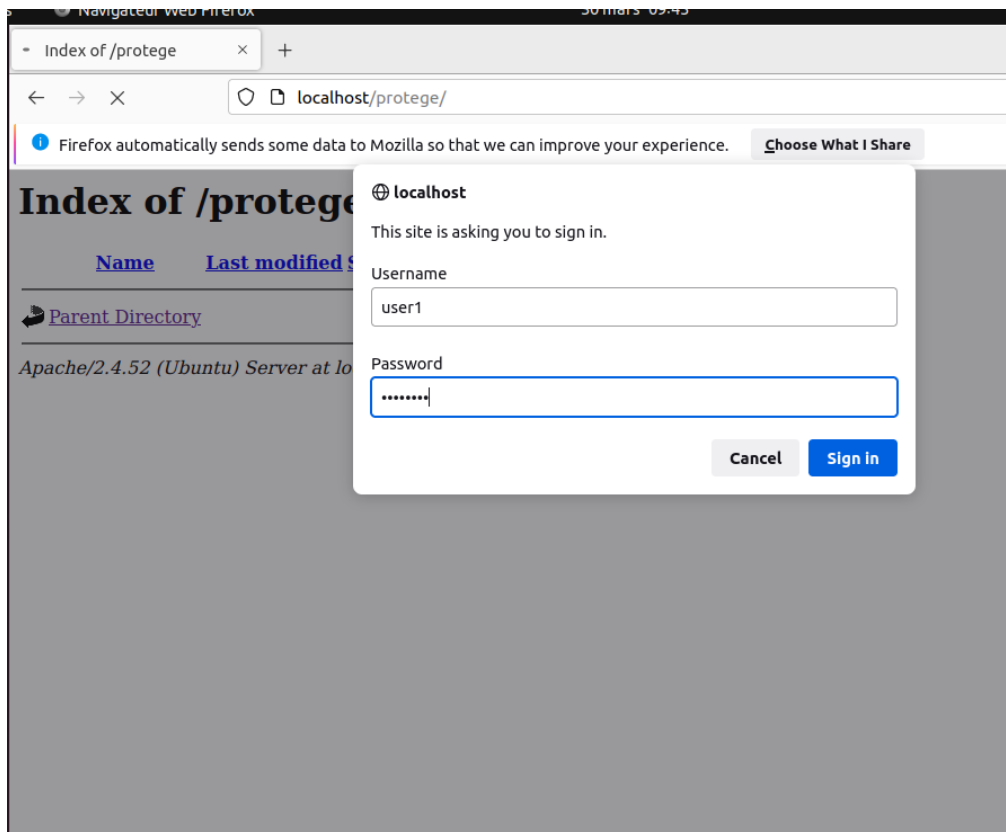
    <Directory "/var/www/html/protège">
        AuthType Basic
        AuthName "Accès réservé au groupe RT1"
        AuthUserFile /etc/apache2/.htpasswd
        AuthGroupFile /etc/apache2/RT1
        Require group RT1
    </Directory>
    # For most configuration files from conf-available/, which are
    # enabled or disabled at a global level, it is possible to
    # include a line for only one particular virtual host. For example the
    # following line enables the CGI configuration for this host only
    # after it has been globally disabled with "a2disconf".
    #Include conf-available/serve-cgi-bin.conf
</VirtualHost>

# vim: syntax=apache ts=4 sw=4 sts=4 sr noet
```

*

Faire un sudo systemctl reload après avoir effectué les modifications

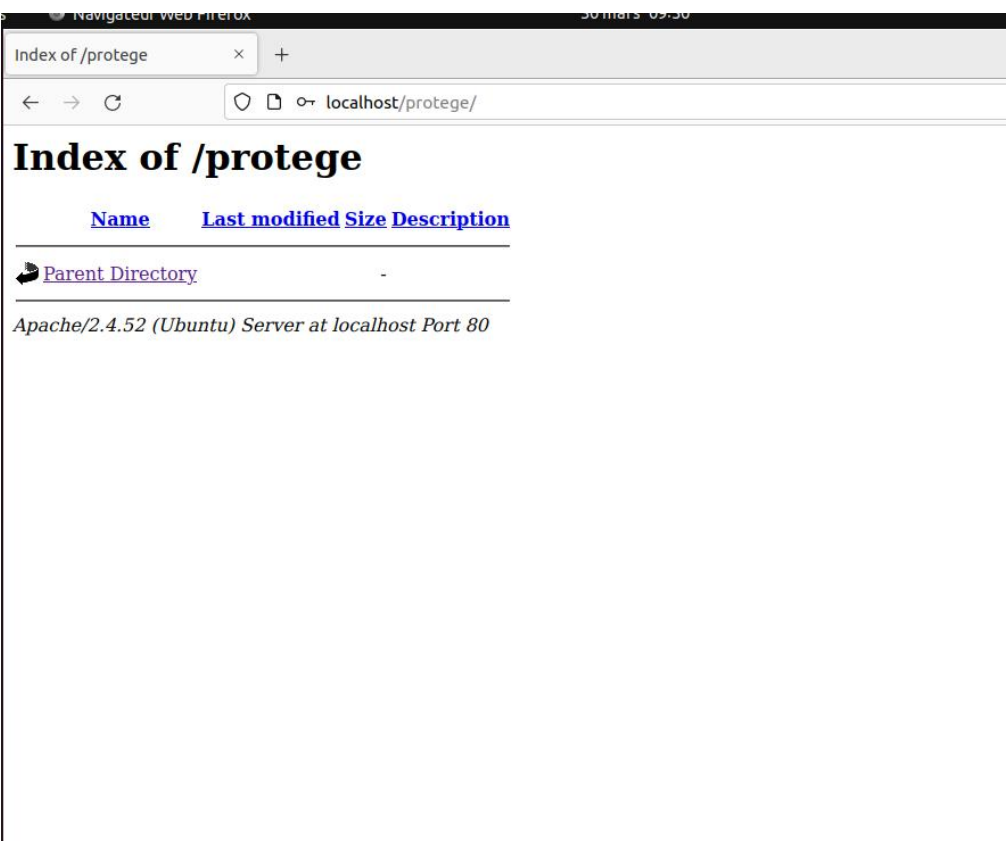
Essai de connexion



Connexion établie

*

Essai de



connexion avec un mauvais nom d'utilisateur et un mauvais mdp

- tail -f /var/log/apache2/error.log

```
[Mon Mar 30 09:47:52.621485 2026] [auth_basic:error] [pid 5302:tid 140058480834112] [client 127.0.0.1:48256] AH01617:
user user2: authentication failure for "/protege/": Password Mismatch
[Mon Mar 30 09:49:22.464568 2026] [auth_basic:error] [pid 5301:tid 140058455656000] [client 127.0.0.1:46028] AH01618:
user user3 not found: /protege/
```

*

Limitation de la bande passante

Activation du module ratelimit

- sudo a2enmod ratelimit
- sudo systemctl restart apache2

Configuration du fichier /etc/apache2/000-default.conf et limitation du débit à 40kbits/s

```
<Directory>
    <Location "/">
        SetOutputFilter RATE_LIMIT
        SetEnv rate-limit 40
        SetEnv no-gzip 1
    </Location>
    # For most configuration files from conf-available/, which are
    # enabled or disabled at a global level, it is possible to
    # include a line for only one particular virtual host. For example the
    # following line enables the CGI configuration for this host only
    # after it has been globally disabled with "a2disconf".
    #Include conf-available/serve-cgi-bin.conf
</VirtualHost>
```

Création d'un fichier test

- sudo dd if=/dev/urandom of=/var/www/html/test_debit.bin bs=1M count=1

Mésurer les débits

- wget http://localhost/test_debit.bin -O /dev/null (depuis une autre machine ou en local)

Test et validation

```
root@rt-mv:~# systemctl restart apache2
root@rt-mv:~# wget http://localhost/test_debit.bin -O /dev/null
--2026-03-30 10:47:57-- http://localhost/test_debit.bin
Résolution de localhost (localhost)... 127.0.0.1
Connexion à localhost (localhost)|127.0.0.1|:80... connecté.
requête HTTP transmise, en attente de la réponse... 200 OK
Taille : 1048576 (1,0M) [application/octet-stream]
Enregistre : '/dev/null'

/dev/null 100%[=====] 1,00M 37,8KB/s ds 26s
2026-03-30 10:48:23 (39,6 KB/s) - '/dev/null' enregistré [1048576/1048576]
root@rt-mv:~#
```

Pages personnelles

Activation du module userdir

- sudo a2enmod userdir
- sudo systemctl restart apache2

```
root@rt-mv:~# a2enmod userdir
Enabling module userdir.
To activate the new configuration, you need to run:
  systemctl restart apache2
root@rt-mv:~# systemctl restart apache2
root@rt-mv:~#
```

Préparer l'espace utilisateur

Créer un dossier

- mkdir ~/public_html

Créer un page de test

- echo « <h1>Ma page perso</h1> » > ~/public_html/index.html

Régler les permissions :

- chmod 755 /home/user
- chmod 755 ~/public_html

```
root@rt-mv:~# a2enmod userdir
Enabling module userdir.
To activate the new configuration, you need to run:
  systemctl restart apache2
root@rt-mv:~# systemctl restart apache2
root@rt-mv:~# mkdir ~/public_html
root@rt-mv:~# echo "<h1>Ma page perso</h1>" > ~/public/html/index.html
> Bonjour et bienvenue
> ^C
root@rt-mv:~# echo "<h1>Ma page perso</h1>" > ~/public/html/index.html
Bonjour et bienvenue
> ^C
root@rt-mv:~# chmod 755 /home/user
chmod: impossible d'accéder à '/home/user': Aucun fichier ou dossier de ce type
root@rt-mv:~# chmod 755 /home/administrateur
root@rt-mv:~# chmod 755 ~/public_html
root@rt-mv:~# nano /etc/apache2/mods-enabled/userdir.conf
root@rt-mv:~# systemctl restart apache2
```

Test et validation



Serveurs virtuels

Création de l'arborescence des sites

- sudo mkdir -p /var/www/vendeur

- sudo mkdir -p /var/www/client

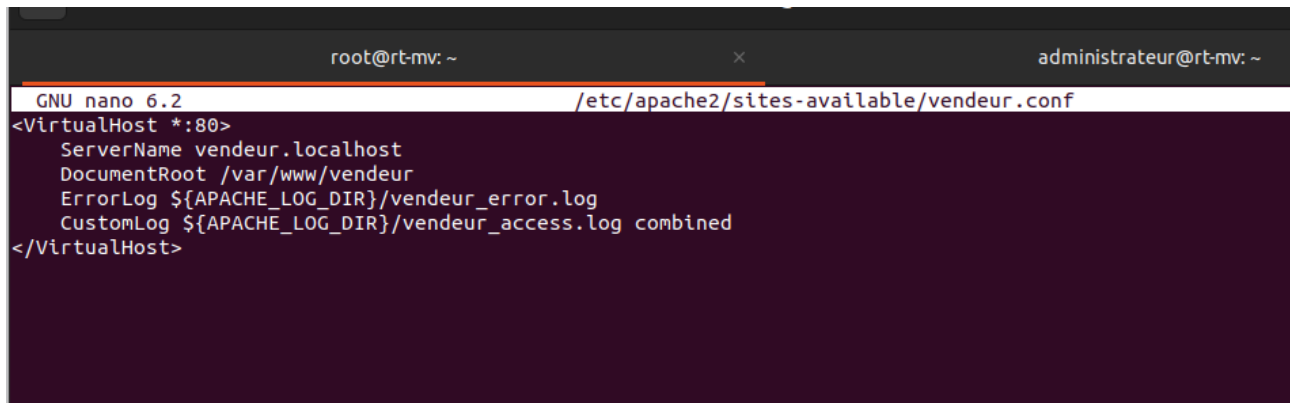
Création des pages web des utilisateurs vendeurs et client

echo "<h1>Site Vendeur</h1>" | sudo tee /var/www/vendeur/index.html

echo "<h1>Site Client</h1>" | sudo tee /var/www/client/index.html

Création des fichiers de configuration Apache

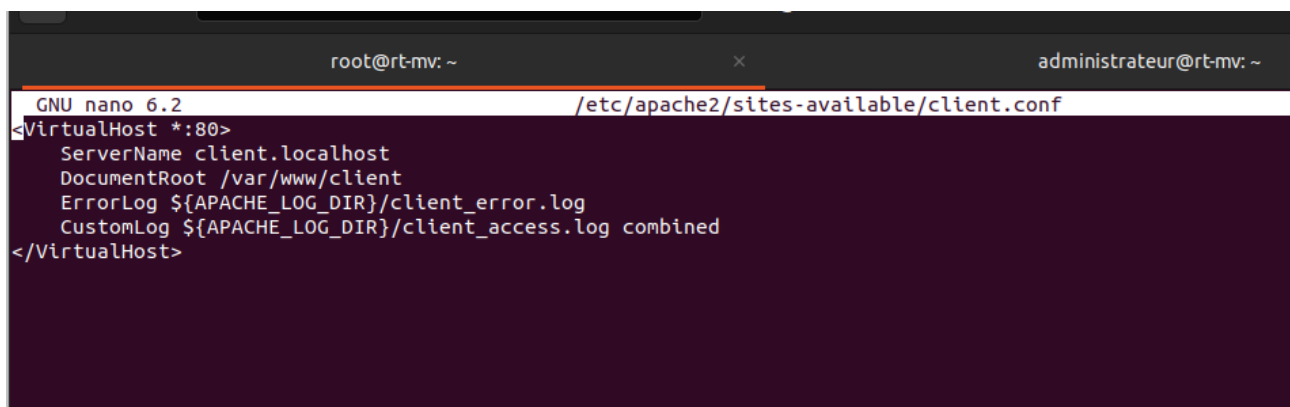
Vendeur



```
root@rt-mv: ~                               administrateur@rt-mv: ~
GNU nano 6.2                                /etc/apache2/sites-available/vendeur.conf
<VirtualHost *:80>
  ServerName vendeur.localhost
  DocumentRoot /var/www/vendeur
  ErrorLog ${APACHE_LOG_DIR}/vendeur_error.log
  CustomLog ${APACHE_LOG_DIR}/vendeur_access.log combined
</VirtualHost>
```

*

Client



```
root@rt-mv: ~                               administrateur@rt-mv: ~
GNU nano 6.2                                /etc/apache2/sites-available/client.conf
<VirtualHost *:80>
  ServerName client.localhost
  DocumentRoot /var/www/client
  ErrorLog ${APACHE_LOG_DIR}/client_error.log
  CustomLog ${APACHE_LOG_DIR}/client_access.log combined
</VirtualHost>
```

*

Activation des sites et relancer

- sudo a2ensite vendeur.conf client.conf (activer les sites)

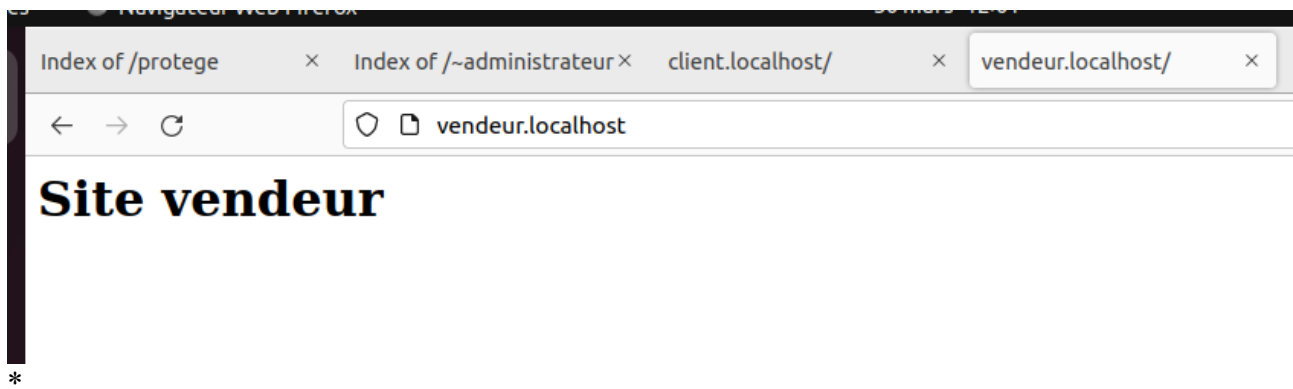
- sudo systemctl restart apache2

Configuration du fichier /etc/hosts (OPTIONNEL)

- sudo nano etc/hosts

127.0.0.1 client.localhost vendeur.localhost

Vérification



2. Machine LEMP